Architecture Selection: Elective Bidding and Allocation System

Lab 3: Component Modelling and Architectural Pattern Selection | PES2UG24CS068

We chose Microservices Architecture for the Academic Elective Bidding and Allocation System.

Reason 1: Two workloads with opposite shapes

NFR-002 requires 2,000 concurrent users at under 3 seconds per action, and that load arrives in the final hours before a round closes. NFR-001 requires the solver to place 5,000 bids in under 30 seconds, which is one long CPU bound batch job. Microservices let us run many stateless replicas of the Bid Management Service during the rush while the Allocation Engine Service runs on separate CPU optimised nodes, so the interactive path and the solver never compete for the same resources. A layered or client server design ships both in a single deployable unit and cannot scale one without the other.

Reason 2: A low priority path must not stop bidding

Notification is Medium priority (FR-004) while bid submission and editing are High (FR-001, FR-002). Because the Notification Service is a separate component reached through a message queue, a mail server outage only delays alerts and bids keep being accepted. In the same way, a fault in the Allocation Engine Service cannot bring down the endpoints students are still using, which matters most at the deadline when there is no time to recover.

Security Advantage

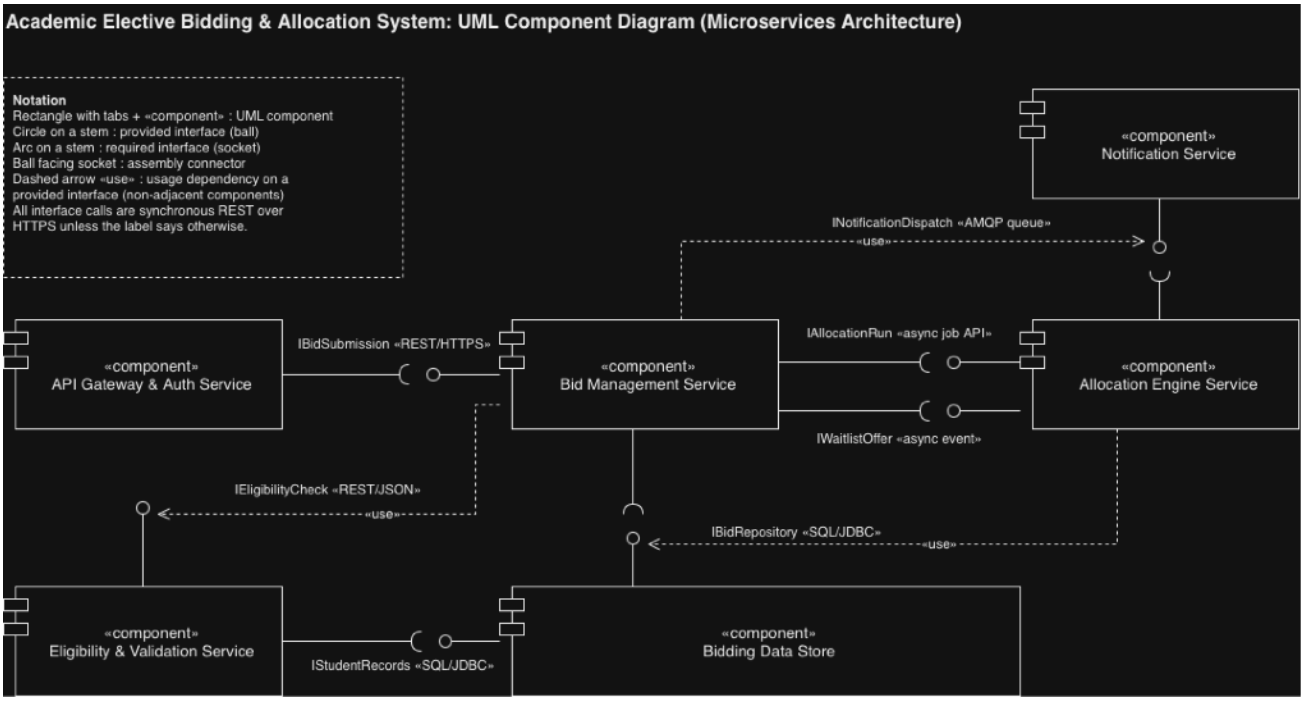
The API Gateway and Auth Service is the only component with a public route. It terminates TLS, verifies the caller's signed token, and applies role based rules before forwarding any call. The Eligibility and Validation Service, Allocation Engine Service and Bidding Data Store sit on a private network with no inbound route from the internet, and services prove their identity to each other with mutual TLS, so a stolen student token is rejected at the service boundary rather than at the database. Each service holds its own database account scoped to its own schema, so the Bid Management Service can write bids and credit balances but cannot write seat capacity or final allocations. The administrator override in UC-05 is a distinct route that requires an administrator role claim and writes an append only audit record of who changed which allocation. The result is a small blast radius: compromising the student facing endpoint does not give an attacker the ability to bypass prerequisites (FR-001) or exceed capacity (FR-003).

Performance Benefit

The allocation run is started through IAllocationRun and executes as an asynchronous job, so the 30 second solve never sits inside a student HTTP request. Submit, edit and withdraw calls keep meeting the 3 second target in NFR-002 even while a round is being solved. Read heavy traffic such as the course catalogue, remaining seats and prerequisite status is served by replicas of the Bid Management Service backed by a cache, and those replicas can be added on their own without resizing the transactional bid store.

Disadvantages Accepted

1. Infrastructure cost and management effort. For an application of this size the container platform, service registry, gateway, message broker and per service monitoring cost more in money and staff time than one deployable application would. The institution pays year round for elasticity it needs for only a few hours per bidding round.
2. Consistency is harder to hold. Credit deduction, seat allocation and waitlist ranking (FR-005) span more than one service, so the design needs idempotent handlers and compensating actions in place of a single database transaction, and there are brief windows where a student sees a recorded bid before the waitlist position appears.


Academic Elective Bidding & Allocation System: UML Component Diagram (Microservices Architecture)
Notation
Rectangle with tabs + «component» : UML component
Circle on a stem : provided interface (ball)
Arc on a stem : required interface (socket)
Ball facing socket : assembly connector
Dashed arrow «use» : usage dependency on a provided interface (non-adjacent components)
All interface calls are synchronous REST over HTTPS unless the label says otherwise.
«component»
Notification Service
INotificationDispatch «AMQP queue»
«use»
«component»
API Gateway & Auth Service
IBidSubmission «REST/HTTPS»
«component»
Bid Management Service
IAllocationRun «async job API»
«component»
Allocation Engine Service
IWaitlistOffer «async event»
IEligibilityCheck «REST/JSON»
«use»
IBidRepository «SQL/JDBC»
«use»
«component»
Eligibility & Validation Service
IStudentRecords «SQL/JDBC»
«component»
Bidding Data Store